



# Simulating and Investigating Long-Lived Axion-Like Particles at the Proposed FCC-ee

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# Abstract

Much of modern high-energy physics is concerned with finding particles not included in the current accepted Standard Model (SM) of particle physics. This report focuses on one such proposed Beyond the Standard Model (BSM) particle, the Axion-Like Particle (ALP). Specifically, this report examines the possibilities of exploring long-lived ALPs at the proposed Future Circular Collider (FCC).

Through theoretical background, Monte Carlo generation, and physics analysis, this report walks through the steps of what potential ALP searches at the FCC would look like. First, an introduction to the current theoretical framework that produces ALPs is given. Next, events are generated using Monte Carlo generators, both for a signal process with various ALP masses and coupling constants and for multiple background processes. These events are then analyzed using the existing FCC physics analysis framework, which is extended to include additional variables and broaden the investigation. An analysis of the lifetimes of these long-lived ALPs is carried out, and they are compared to analytical computations. This is followed by an overview of potential background processes with similar signatures and suggestions on multiple selections that could be implemented to isolate the signal process. Finally, a sensitivity analysis of the FCC-ee to the signal process was completed, giving an idea of which ALP masses and coupling constants are most sensitive to detection.

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# Abbreviations

**ALP** Axion-Like Particle

**BSM** Beyond the Standard Model

**CERN** European Council for Nuclear Research

**CP** Charge-Parity

**DCH** Drift Chamber

**DESY** Deutsches Elektronen-Synchrotron

**FCC** Future Circular Collider

**FCC-ee** Future Circular Collider–electron-positron

**FCC-eh** Future Circular Collider–electron-hadron

**FCC-hh** Future Circular Collider–hadron-hadron

**IDEA** Innovative Detector for Electron-positron Accelerators

**LHC** Large Hadron Collider

**LLP** Long-Lived Particle

**MC** Monte Carlo

**MIT** Massachusetts Institute of Technology

**SM** Standard Model

**VTX** Silicon-Vertex Tracker

# 1 Introduction

The experimental confirmation of the existence of the Higgs Boson occurred in 2012, at the Large Hadron Collider (LHC), and was the final step in the completion of what is known as the Standard Model (SM) of particle physics [1]. The SM has been verified experimentally with a high degree of precision and is today’s most complete model of particle physics. However, despite its scope and accuracy, there are multiple phenomena that are not explained by the SM. Among these are the existence of dark matter, the baryonic asymmetry of the universe, and the existence of neutrino masses. In order to explain these phenomena, physicists theorize particles Beyond the Standard Model (BSM).

Multiple proposed particles exist in BSM theories, but this report will focus on Axion-Like Particles (ALPs). ALPs are hypothesized light pseudoscalar particles with a wide range of potential masses and coupling constants [2]. For this report, the ALPs of note fall under the category of Long-Lived Particles (LLPs), meaning their lifetimes are often longer than typically analyzed SM particles, and thus they can be more difficult for detectors to reconstruct. LLPs are important for BSM searches since a lot of SM background is not long-lived, making it potentially easier to notice the desired signal.

The Axion itself was originally proposed in 1977 as a solution for the strong Charge-Parity (CP) problem [3]. ALPs can be produced in a variety of processes and can decay to a variety of particles, but this report will focus on a single ALP produced in association with a photon, after which the ALP decays into a pair of photons (see Figure 2).

In order to further investigate BSM theories beyond the LHC, physicists suggest multiple new particle colliders to be built in the future. This report will focus on the Future Circular Collider (FCC), proposed to be the largest particle accelerator in the world, to be built at CERN with a circumference of approximately 100 km (compared to the 27 km circumference of the LHC) [4]. The FCC is planned to contain multiple stages, beginning with electron-positron collisions (FCC-ee) before moving on to electron-hadron (FCC-eh) and hadron-hadron (FCC-hh) collisions. This report is concerned with the former.

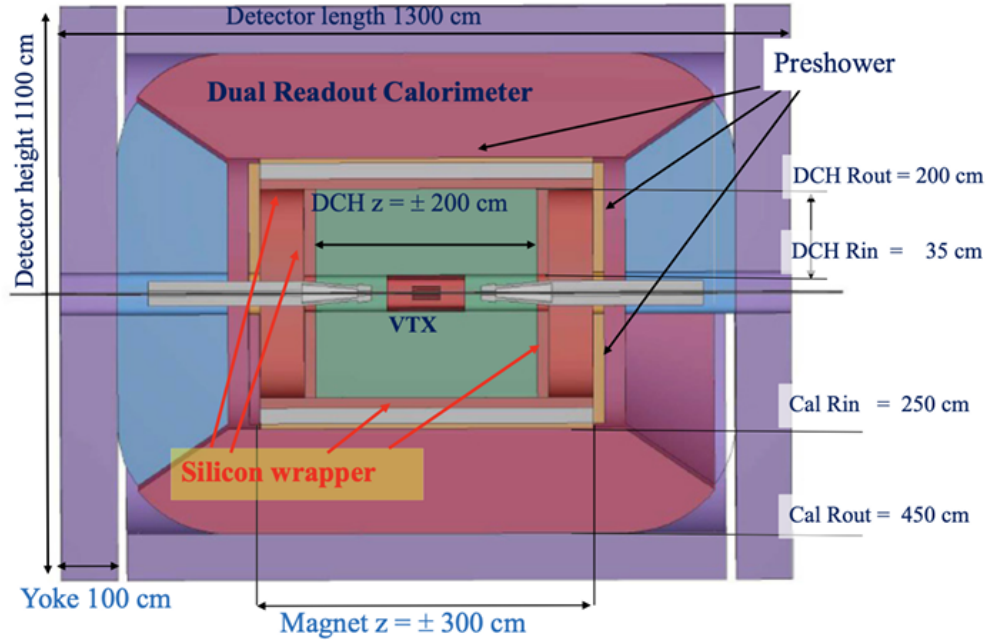
The aim of the analysis outlined in this report is to simulate relevant Monte Carlo (MC) events and investigate the properties of long-lived ALPs at the FCC-ee. This includes manipulating variables such as ALP mass and coupling constant, looking at relevant backgrounds, and proposing potential selections to eliminate or reduce these background events.

## 1.1 FCC-ee

The FCC is proposed to become the largest particle accelerator in the world, and its first stage, the FCC-ee, is planned to provide high-energy electron-positron collisions. The goal of building the FCC is to explore BSM theories and gain experimental evidence for new hypothesized particles, such as the ALP. This analysis is concerned with the Z pole run of the FCC-ee, which will occur at a center of mass energy  $\sqrt{s} = 91.0 \text{ GeV}$ .

The potential construction of the FCC is still far in the future, yet a timeline exists for planned stages to reach eventual completion. First, the FCC must undergo a feasibility study, which is currently in progress and will measure issues such as funding and resources against the capability of the collider to answer unresolved physics questions [5]. As part of this feasibility study, research is being done using Monte Carlo simulations of electron-positron collisions and reconstructing processes through simulations of proposed detectors, which would be placed at certain points along the FCC, similarly to the LHC.

The proposed detector concept that is relevant for this report is the Innovative Detector for Electron-positron Accelerators (IDEA). A diagram of the IDEA detector can be seen in Figure 1. It is applicable to this analysis that, in high-energy physics experiments, the produced photons are detected only by the electromagnetic calorimeter. However, electrons and positrons are also detected by the electromagnetic calorimeter, meaning careful analysis is necessary to separate photons and electrons/positrons in reconstruction.



**Figure 1:** A diagram of the proposed IDEA detector (acronyms defined in the Abbreviations section). The dual readout calorimeter is what detects reconstructed photons [4].

## 2 Theory

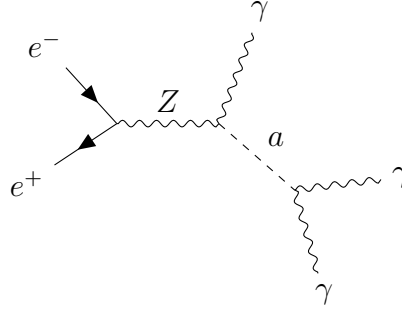
The theory section of this report will provide background material, explain the context of this analysis, and serve as a framework to set up the equations used later in the analysis of the experimental data.

### 2.1 Signal

Although there are many processes that have the potential to generate an ALP, this report is concerned with the process where the electron-positron pair forms a Z boson (which would be the most likely decay mode during the Z-pole run of the FCC-ee, which this report is focused on), which decays into a photon and an ALP, which then further decays into two photons:

$$\text{Signal process: } e^+e^- \rightarrow Z \rightarrow a\gamma, \text{ where } a \rightarrow \gamma\gamma \quad (1)$$

The Feynman diagram for this process can be seen in Figure 2.



**Figure 2:** Relevant Feynman diagram for the signal process.

For all processes, an integrated luminosity  $\mathcal{L} = 150 \text{ ab}^{-1}$  is assumed. For the generated ALP, the mass  $m_a$  and the coupling constant to photons  $C_{\gamma\gamma}$  were varied.

ALP interactions with SM particles up to dimension-5 can be described by the following Lagrangian [6]:

$$\begin{aligned} \mathcal{L}_{\text{eff}} = & \frac{1}{2}(\partial_\mu a)(\partial^\mu a) - \frac{m_a^2}{2}a^2 + \sum_f \frac{c_{ff}}{2} \frac{\partial^\mu a}{\Lambda} \bar{f} \gamma_\mu \gamma_5 f \\ & + g_s^2 C_{GG} \frac{a}{\Lambda} G_{\mu\nu}^A \tilde{G}^{\mu\nu,A} + g^2 C_{WW} \frac{a}{\Lambda} W_{\mu\nu}^A \tilde{W}^{\mu\nu,A} + g'^2 C_{BB} \frac{a}{\Lambda} B_{\mu\nu} \tilde{B}^{\mu\nu} . \end{aligned} \quad (2)$$

### 2.2 ALP Lifetime

As this analysis is interested in studying long-lived ALPs, it is essential to understand the lifetime of ALPs within the detector. One can start by looking at decay width ( $\Gamma$ ), defined in the following equation [7]:

$$\Gamma(a \rightarrow \gamma\gamma) = \frac{4\pi\alpha^2 m_a^3}{\Lambda^2} |C_{\gamma\gamma}^{\text{eff}}|^2 , \quad (3)$$

where  $\alpha$  is the strength of the electromagnetic force, set to  $\frac{1}{127.9}$  in the ALP model used,  $m_a$  is the mass of the ALP,  $\Lambda$  is the global symmetry breaking scale, and  $C_{\gamma\gamma}^{\text{eff}}$  is the effective coupling constant of the ALP to photons. For the ALP masses relevant to this analysis ( $m_a > \Lambda_{QCD}$ ),  $C_{\gamma\gamma}^{\text{eff}}$  satisfies the following equation:

$$C_{\gamma\gamma}^{\text{eff}}(m_a \gg \Lambda_{QCD}) = C_{\gamma\gamma} + \sum_f \frac{N_c^f Q_f^2}{16\pi^2} c_{ff} B_1(\tau_f) + \frac{2\alpha}{\pi} \frac{C_{WW}}{s_w^2} B_2(\tau_W). \quad (4)$$

In the model used for this analysis, however,  $c_{ff}$  and  $C_{WW}$  are set to 0 (Appendix A), leaving  $C_{\gamma\gamma}^{\text{eff}}$  equal to  $C_{\gamma\gamma}$ .

In addition, the lifetime of the ALP ( $\tau$ ) is inversely proportional to the decay width  $\Gamma$ , satisfying the following relationship [8]:

$$\tau(a \rightarrow \gamma\gamma) = \frac{\hbar}{\Gamma(a \rightarrow \gamma\gamma)}, \quad (5)$$

where  $\hbar$  is the reduced Planck's constant ( $\frac{h}{2\pi}$ ).

Finally, the global symmetry breaking scale  $\Lambda$  can be defined in terms of  $f_a$ , a parameter used in the event simulations to specify the new physics scale (in this case,  $f_a = \frac{1000}{16\pi^2} \text{ GeV} \approx 6.33 \text{ GeV}$ ) [7]:

$$\Lambda = 32\pi^2 f_a |C_{GG}^{\text{eff}}|, \quad (6)$$

where  $C_{GG}^{\text{eff}}$  is the effective coupling constant of the ALP to gluons and is set to 0.5 in the model.

Combining equations (3), (4), (5), and (6), the ALP lifetime for this analysis is defined as:

$$\tau(a \rightarrow \gamma\gamma) = \frac{64\pi^3 \hbar f_a^2}{\alpha^2 m_a^3 C_{\gamma\gamma}^2}. \quad (7)$$

## 2.3 Special Relativity

Special relativity concepts are essential to high-energy particle physics, and for this report, they will be utilized to examine different reference frames in which ALP lifetime can be measured. The difference in reference frames in the high-energy scenarios relevant to this report is represented by the Lorentz factor  $\gamma$ , which, according to relativistic mass transformation, can be represented as a ratio between an object's mass in an inertial reference frame ( $m$ ) and its rest mass in a non-inertial reference frame ( $m_0$ ) [9]:

$$m = \gamma m_0. \quad (8)$$

Using the basic definition of momentum  $p = mv$ , equation (8) can be utilized to show relativistic momentum:

$$p = mv = \gamma m_0 v. \quad (9)$$

Finally, the Lorentz factor can be isolated, and one can utilize the fact that in the particle physics scenario relevant to this report,  $v = c$ , the speed of light (which has

been set in high-energy particle physics to  $c = 1$  in order to simplify computations). Solving this through and applying it to ALPs yields the following:

$$\gamma = \frac{p}{m_0 v} = \frac{p}{m_0 c} = \frac{p}{m} = \frac{p_a}{m_a}, \quad (10)$$

where  $p_a$  and  $m_a$  represent the momentum and mass of the ALP, respectively.

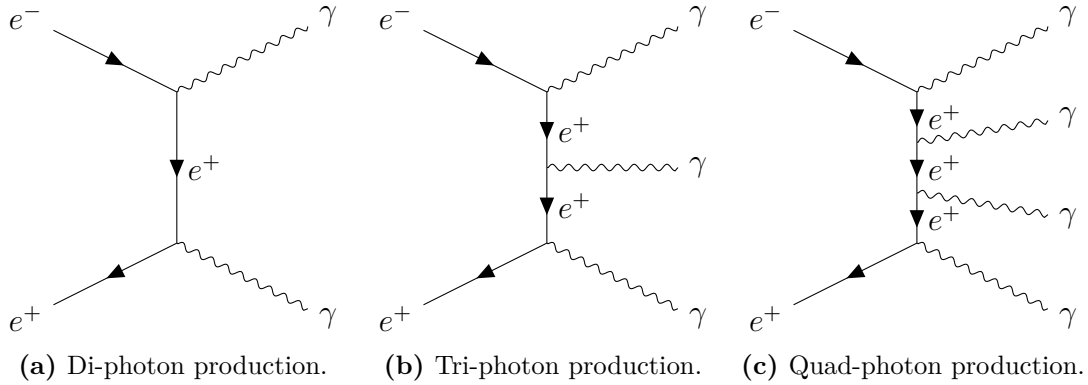
## 2.4 Backgrounds

While the signal for this analysis is the process represented in Figure 2, multiple backgrounds have also been studied with signatures that may appear similar to the signal in reconstruction. The backgrounds investigated in this report are di-, tri-, and quad-photon production (Figure 3):

$$\text{Di-photon production: } e^+e^- \rightarrow \gamma\gamma \quad (11)$$

$$\text{Tri-photon production: } e^+e^- \rightarrow \gamma\gamma\gamma \quad (12)$$

$$\text{Quad-photon production: } e^+e^- \rightarrow \gamma\gamma\gamma\gamma \quad (13)$$



**Figure 3:** Feynman diagrams for background processes.

## 2.5 Normalization

Throughout event generation, a certain number of total events (100000 or 10000) was simulated. However, when considering event selection or the sensitivity the FCC-ee will have to certain processes, it can be more informative to look at the expected number of events which will occur. For this report, the Z pole run at the FCC-ee is considered, with an integrated luminosity of  $\mathcal{L} = 150 \text{ ab}^{-1}$ . To normalize the raw number of events from simulation and compute the expected number of events ( $n$ ), the following equation is used [10]:

$$n = \frac{k\epsilon\sigma\mathcal{L}}{N_{tot}} n_{raw}, \quad (14)$$

where the  $k$  factor and the matching efficiency  $\epsilon$  are set to one in this report,  $\sigma$  represents the cross-section of the process,  $N_{tot}$  is the total number of events (100000 or 10000),

and  $n_{raw}$  is the number of events after selection(s) are made. This value  $n$  can then be used in order to calculate the sensitivity ( $s$ ) of the FCC-ee Z pole run to the signal process based on relevant background(s) [10]:

$$s = \frac{n_S}{\sqrt{n_S + n_B}}, \quad (15)$$

where  $n_S$  is the expected number of signal events and  $n_B$  is the expected number of background events.

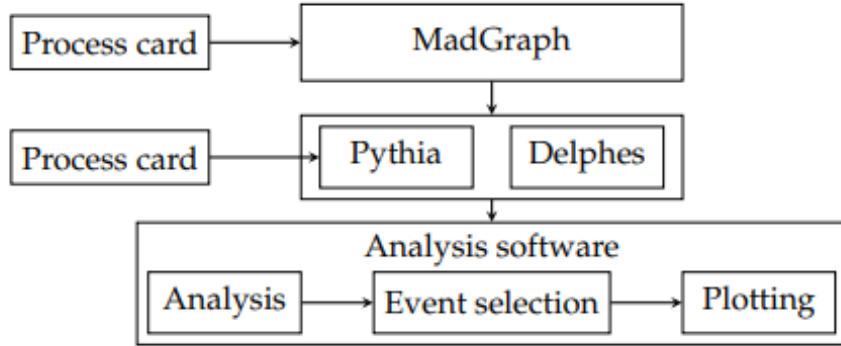
### 3 Methodology

Due to the fact that the FCC has not yet been built, no experimental data was available for this analysis. Instead, Monte Carlo event generation was used, followed by simulated detector reconstruction, then followed by analysis scripts that were used to plot the generated data as well as make selections and introduce new variables.

#### 3.1 Simulation

Many centrally produced samples of FCC-ee collision events exist, both from the Spring 2021 and the Winter 2023 campaigns [11], but for this analysis, new event samples were privately generated\* to ensure consistency and up-to-date simulation.

First, a process card, specifying the desired process (either signal or background) as well as other variables (all coupling constants except  $C_{\gamma\gamma}$  are set to zero,  $f_a = 6.33^\dagger$  (6), and  $m_a$  and  $C_{\gamma\gamma}$  are varied throughout this analysis) (Appendix A), is used in conjunction with *MadGraph5\_aMC@NLO* version 3.5.4 in order to generate Monte Carlo events [12]. Next, via another process card (Appendix B), the programs Pythia 8 and Delphes are run simultaneously in order to simulate the parton shower and the IDEA detector reconstruction, respectively [13][14]. The output file format is EDM4hep, which can be analyzed using Python/PyROOT scripts [15][16]. This entire process can be visualized in Figure 4.



**Figure 4:** Diagram showing the simulation and analysis process [10].

#### 3.2 Analysis

Once the events are produced with MadGraph, Pythia 8, and Delphes, physics analysis is performed on the resulting ROOT file. For this, the FCCAnalyses software, an ongoing project used for performing physics analyses relevant to the FCC, was used and extended through the addition of useful variables (examples given in subsection 4.2

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\*All generated samples located on CERN's LXPLUS service: `/eos/experiment/fcc/ee/analyses_storage/BSM/LLPs/ALPs_3photons/ALP_sample_creation/`.

<sup>†</sup>This is notable because the value of  $f_a$  in the MadGraph process card was updated in this report to 6.33 rather than 1000, which was the value used in previous papers.

and subsection 4.3)<sup>‡</sup> [17].

Three separate Python scripts within the FCCAnalyses software framework were used for this analysis. The first of these, “stage 1”, is used to define variables based on the outputted event data, often referencing EDM4hep C++ scripts elsewhere within FCCAnalyses [15]. The next stage, named “final”, defines the parameters for the plots being created, as well as carries out any event selections desired to minimize background. The final step in the process, “plots”, is responsible for depicting the processed data in histograms.

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<sup>‡</sup>The fork with all added and updated variables: <https://github.com/merlinothefirst/FCCAnalyses>.

## 4 Results

Using the aforementioned methodology, plots and tables have been created to conveniently visualize the outcomes of various collision processes. These plots have been analyzed in order to show trends and patterns. In some cases, selections have been made in an attempt to reduce background. Additionally, preexisting theory has been utilized to analytically calculate certain values for comparison. In the following histograms, any events associated with pure simulation (MadGraph + Pythia 8) are labeled as “gen” level, and any events associated with detector reconstruction (Delphes) are labeled as “reco” level. All distributions without backgrounds have been normalized to one, and all distributions with backgrounds have been normalized to the expected number of events (Equation 14). For all plots, 100000 events were generated (with the exception of subsection 4.4), with an assumed integrated luminosity  $\mathcal{L} = 150 \text{ ab}^{-1}$  and a center of mass energy  $\sqrt{s} = 91.0 \text{ GeV}$ .

### 4.1 Lifetime

As was previously considered in the theory section of this report (section 2), investigating the ALP lifetime is an important aspect of understanding the behavior of these potential particles. The lifetime can help us determine whether an ALP is long-lived, whether it decays into two photons almost instantaneously, or whether it escapes detection entirely.

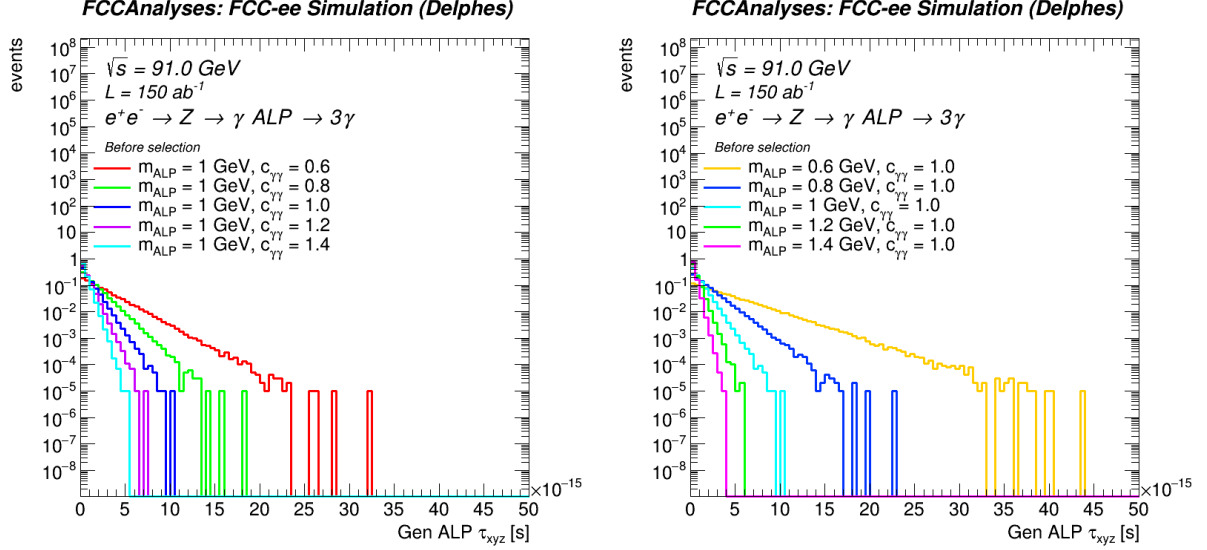
As a first step, the lifetime  $\tau_{xyz}$  is calculated from equations (16) and (17) in “stage 1” of the FCCAnalyses script, and is then plotted (Figure 5) with both varying masses  $m_a$  and varying coupling constants  $C_{\gamma\gamma}$ :

$$L_{xyz} = \sqrt{v_x^2 + v_y^2 + v_z^2}, \quad (16)$$

$$\tau_{xyz} = \frac{L_{xyz} m_a}{10^3 p_a c}, \quad (17)$$

where  $L_{xyz}$  is the distance over which the ALP decays (in  $mm$ ),  $v_x$ ,  $v_y$ , and  $v_z$  represent the respective vertices of one of the photons decaying from the ALP,  $p_a$  is the momentum of the ALP, and  $c$  is the speed of light.

From Figure 5, qualitative relationships can be observed regarding lifetime, coupling constant, and ALP mass. Specifically, it is clear from Figure 5(a) that as the ALP mass remains constant, an increase in coupling constant leads to a decrease in lifetime. Additionally, from Figure 5(b), one can see that as the coupling constant remains constant, an increase in ALP mass leads to a decrease in lifetime. Indeed, these qualitative relationships can be quantified using equation (7), which shows an inverse quadratic relationship between lifetime and coupling constant and an inverse cubic relationship between lifetime and ALP mass. To further prove the consistency between the theoretical equation (7) and the generated plots, Table 1 has been created comparing analytically calculated values vs. the mean of the histograms resulting from Monte Carlo simulation, for a variety of different ALP masses and coupling constants.



(a) Lifetime at ALP mass  $m_a = 1 \text{ GeV}$  and multiple coupling constants  $C_{\gamma\gamma}$ . (b) Lifetime at coupling constant  $C_{\gamma\gamma} = 1.0$  and multiple ALP masses  $m_a$ .

**Figure 5:** ALP lifetimes with varying coupling constants or masses.

| $m_a \text{ (GeV)}$ | $C_{\gamma\gamma}$ | Analytically calculated $\tau_{xyz} \text{ (s)}$ (7) | Mean of MC generated $\tau_{xyz} \text{ graph (s)}$ |
|---------------------|--------------------|------------------------------------------------------|-----------------------------------------------------|
| 1.0                 | 0.6                | 2.378E-15                                            | $2.373\text{E-15} \pm 7.542\text{E-18}$             |
| 1.0                 | 0.8                | 1.338E-15                                            | $1.338\text{E-15} \pm 4.250\text{E-18}$             |
| 1.0                 | 1.0                | 8.561E-16                                            | $8.561\text{E-16} \pm 2.712\text{E-18}$             |
| 1.0                 | 1.2                | 5.945E-16                                            | $5.944\text{E-16} \pm 1.885\text{E-18}$             |
| 1.0                 | 1.4                | 4.368E-16                                            | $4.359\text{E-16} \pm 1.385\text{E-18}$             |
| 0.6                 | 1.0                | 3.964E-15                                            | $3.948\text{E-15} \pm 1.245\text{E-17}$             |
| 0.8                 | 1.0                | 1.672E-15                                            | $1.670\text{E-15} \pm 5.304\text{E-18}$             |
| 1.0                 | 1.0                | 8.561E-16                                            | $8.561\text{E-16} \pm 2.712\text{E-18}$             |
| 1.2                 | 1.0                | 4.954E-16                                            | $4.960\text{E-16} \pm 1.577\text{E-18}$             |
| 1.4                 | 1.0                | 3.120E-16                                            | $3.120\text{E-16} \pm 9.924\text{E-19}$             |

**Table 1:** Lifetime values calculated from the analytical computation vs. from Monte Carlo generation.

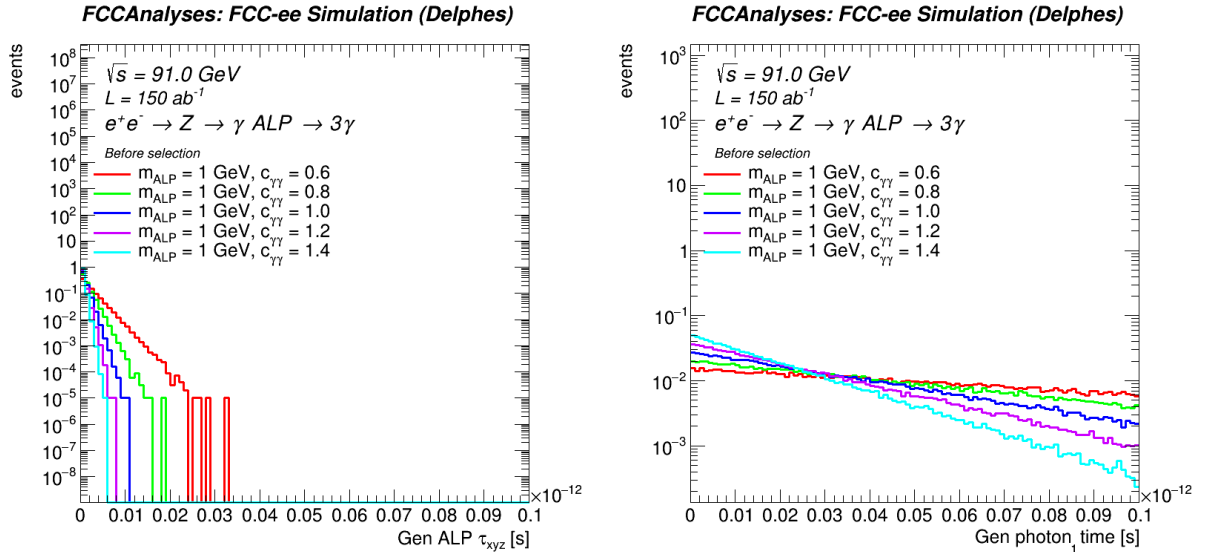
Table 1 shows that the analytically computed results are consistently within the uncertainty of the mean of the generated histogram—with the exception of the data point with  $m_a = 0.6$  and  $C_{\gamma\gamma} = 1.0$ . At this point, the analytical value doesn't fit within the uncertainty of the mean of the histogram. This is likely due to the fact that at these low ALP mass ranges, the lifetime is long enough where some ALPs are escaping the range of the detector, potentially increasing the margin of error of the simulation. Importantly, the new MC generated lifetime values verify the value of  $f_a$  used in this report, which is now 6.33 instead of 1000 (footnote †).

## 4.2 Special Relativity

When considering the variable  $\tau_{xyz}$ , representing the lifetime of the ALP in seconds, it is important to note which reference frame is being considered. Because experiments

at particle accelerators such as the LHC or the FCC are being performed at such high energies, where particles are traveling at almost the speed of light, time dilation due to special relativity plays a role and causes a discrepancy between times in the inertial (moving with the particle) and non-inertial (standing still with respect to the Earth's movement) reference frames.

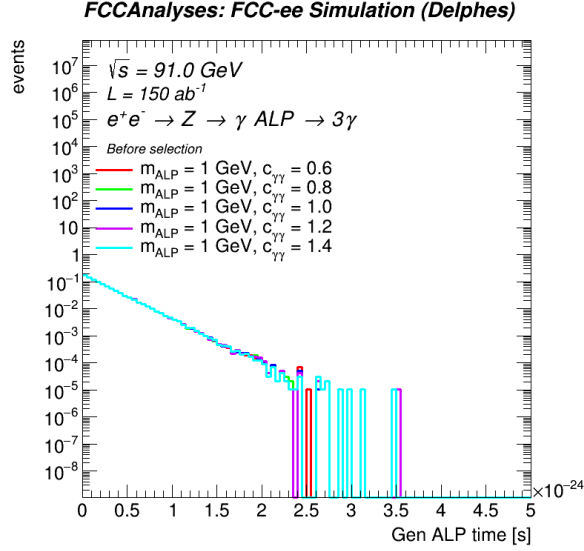
In order to determine the reference frame in which the previously calculated  $\tau_{xyz}$  takes place, another variable, photon1\_time, has been added to the FCCAnalyses framework. This variable represents the time at which one of the two photons decays from the ALP. (photon1\_time and photon2\_time are identical, as they both are a result of the ALP's decay, so for simplicity's sake only photon1\_time is used here.) Naïvely, it seems that photon1\_time should look the same as  $\tau_{xyz}$ , as photon1 decays from the ALP at the same time that the ALP's lifetime has reached its completion. (Notably, the lifetime of the Z boson shown in the Feynman diagram in Figure 2 is essentially negligible on the scales being considered here; graphical proof of this in Figure 7, showing a new variable ALP\_time, the time at which the ALP has decayed from the Z boson, is significantly less than either  $\tau_{xyz}$  or photon1\_time.) However, when plotting this variable on the same set of axes as  $\tau_{xyz}$ , it is clear that the two plots are offset by a certain margin (Figure 6), pointing to a difference in reference frames. To discover the



(a) The previously shown ALP lifetime  $\tau_{xyz}$ , with a constant ALP mass, shown on a different x-axis for better comparison. (b) The newly introduced photon1\_time variable, shown on the same axes as  $\tau_{xyz}$  for comparison.

**Figure 6:**  $\tau_{xyz}$  vs. photon1\_time. Although the slopes for each coupling constant appear consistent, it is clear that the two plots are offset.

exact parameters of this margin of difference, one must consider the difference between the inertial and non-inertial reference frames. This is where special relativity comes into play: as was determined previously in equation (10), the Lorentz factor can be defined, in this case, as  $\frac{p_a}{m_a}$ . (This  $\frac{p_a}{m_a}$  relationship has been seen previously, inversely, in equation (16), as the FCCAnalyses framework is essentially handling the switching of



**Figure 7:** ALP\_time, proving that the lifetime of the Z boson (on the scale of  $10^{-24}$  s) is much shorter than that of the ALP.

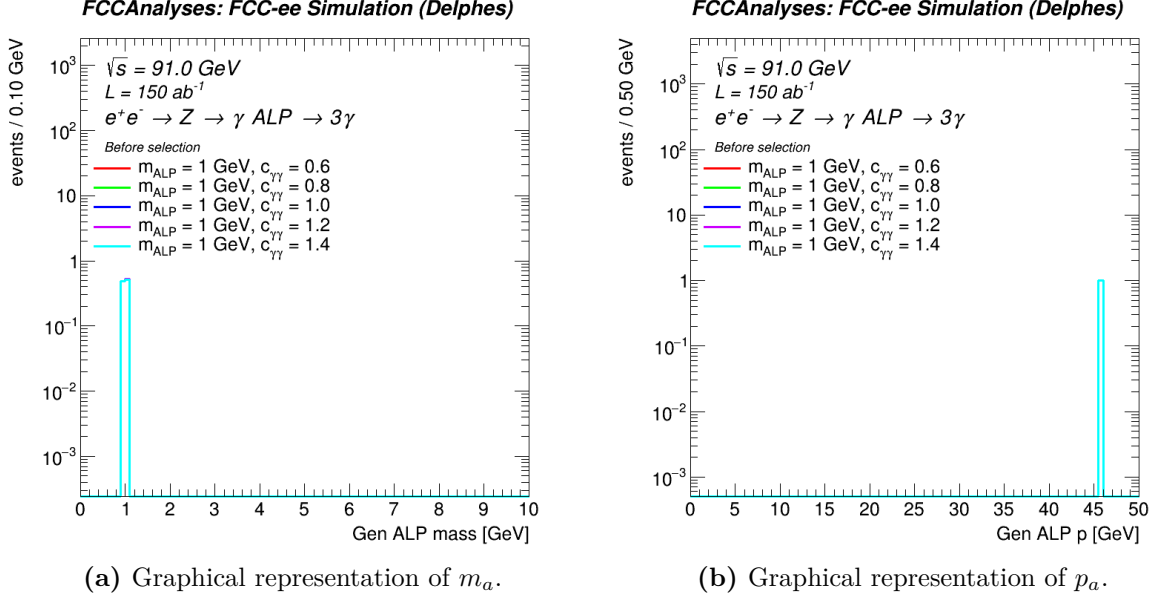
reference frames.) To determine which time variable represents which reference frame, it is necessary to utilize equation (10) to determine the relationship between  $\tau_{xyz}$  and photon1\_time. While  $m_a$  is set to 1 GeV, the momentum  $p_a$  will be exactly half of the center of mass energy  $\sqrt{s} = 91.0$  GeV, or 45.5 GeV (further graphical proof of this can be seen in Figure 8). Thus,  $\frac{p_a}{m_a} = 45.5$ , and using the fact that photon1\_time is greater than  $\tau_{xyz}$  (Figure 6), the following time dilation equation can be used to compare the two different time variables:

$$\text{photon1\_time} = \gamma \tau_{xyz} = \frac{p_a}{m_a} \tau_{xyz} . \quad (18)$$

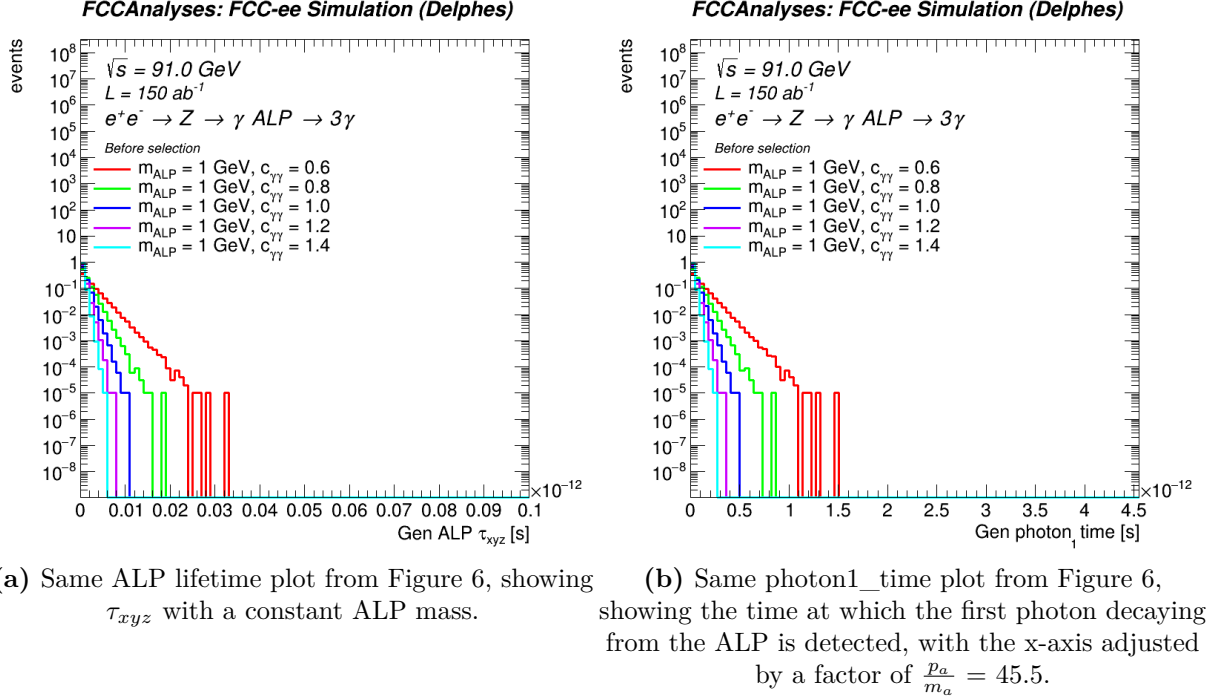
This equation checks out when modifying the x-axis by a factor of  $\frac{p_a}{m_a}$  (Figure 9), including for different masses. To conclude, the relationship represented in equation (10) proves that the variable  $\tau_{xyz}$  represents the ALP lifetime in the inertial reference frame and the variable photon1\_time represents the ALP lifetime (plus the essentially negligible Z boson lifetime (Figure 7)) in the non-inertial reference frame.

### 4.3 Useful Selections

When looking at real data that would be collected from the FCC, it is much more complex than feeding a generator a signal process and analyzing the results. In a real electron-positron collision, there are a variety of different processes that could occur. In order to investigate the ALP specifically, it is important to isolate the signal process shown in Figure 2. However, as mentioned in subsection 2.4, there are multiple relevant background processes (Figure 3) that might have final states that appear similar in reconstruction to the signal. A goal of this project was to find useful selections that would remove many of the background events without removing too many of the signal events. To further investigate these backgrounds and come up with these selections, a few signal points with varying ALP masses and coupling constants were plotted alongside the three applicable backgrounds (Figure 3). Figure 10 shows the number of final



**Figure 8:** ALP mass and momentum. As expected,  $m_a = 1.0 \text{ GeV}$  and  $p_a = 45.5 \text{ GeV}$ .

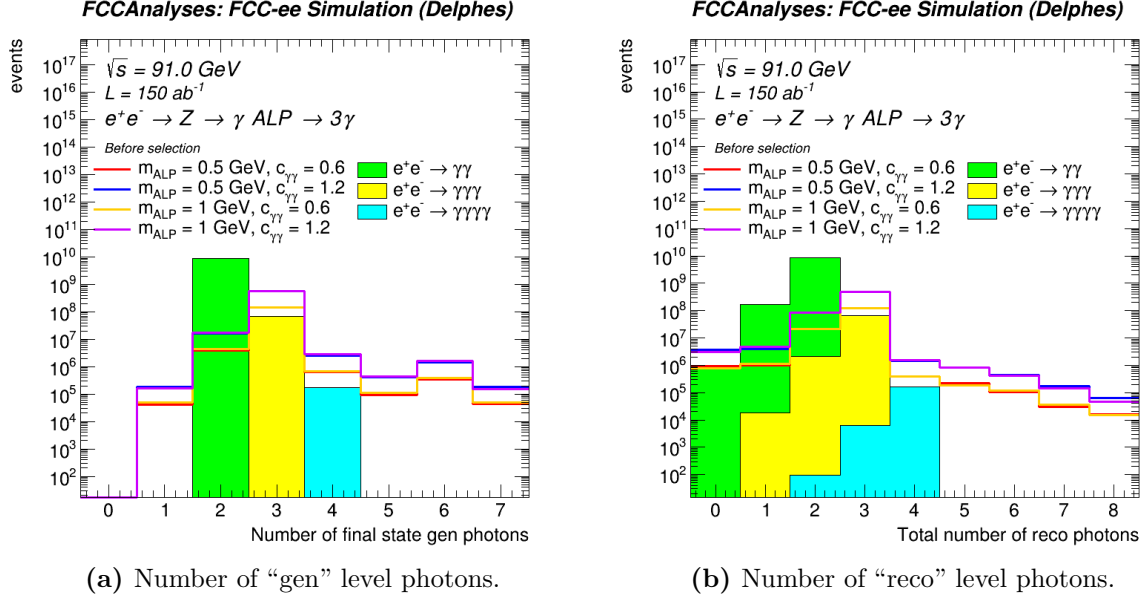


**Figure 9:**  $\tau_{xyz}$  vs. photon1\_time, with x-axis of photon1\_time adjusted to show relationship. This proves that the two variables are off by a factor of 45.5.

state photons, both at the “gen” level and at the “reco” level, for the signal and backgrounds. Notably, signal cases at the “gen” level with a value other than three final state photons are due to Pythia 8 radiating an ALP or a photon more or less than

desired [8].

The logical first selection is to stipulate that there must be exactly three reconstructed



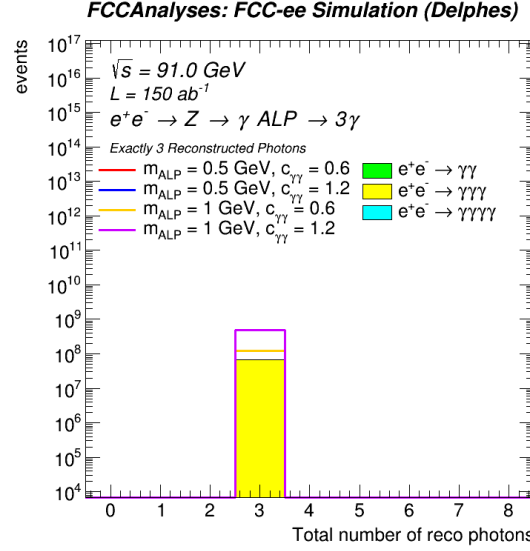
**Figure 10:** “gen” and “reco” level photons. At the “reco” level, imperfections in the detector mean that not all photons are counted in some cases.

photons, as that is the number specified by our signal process (Figure 4). Additionally, based on the plots in Figure 10, this would appear to eliminate all of the di-photon production background (11) and much of the quad-photon production background (13). Indeed, this expected result can be seen in Figure 11, where only the tri-photon production background (12) can be seen. In reality, all of the di-photon production background has been removed as expected; however, a small amount of the quad-photon production background remains, only is so insignificant as to not be visible in comparison to the signal and the tri-photon production background.

The next goal was to greatly eliminate the tri-photon production background. In order to attempt this, one must consider the difference between the signal process and said background. As one can see from examining the Feynman diagram of the signal process (Figure 2), there is a stark difference between the first photon being generated (in conjunction with the ALP), and the second and third photons being generated (both decaying from the ALP). To highlight this difference, new variables  $\Delta R$  and  $\min(\Delta R)$  were added to the FCCAnalyses framework’s analysis scripts. First,  $\Delta R$ , representing the angle between two particles, was calculated using the following equation:

$$\Delta R = \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2}, \quad (19)$$

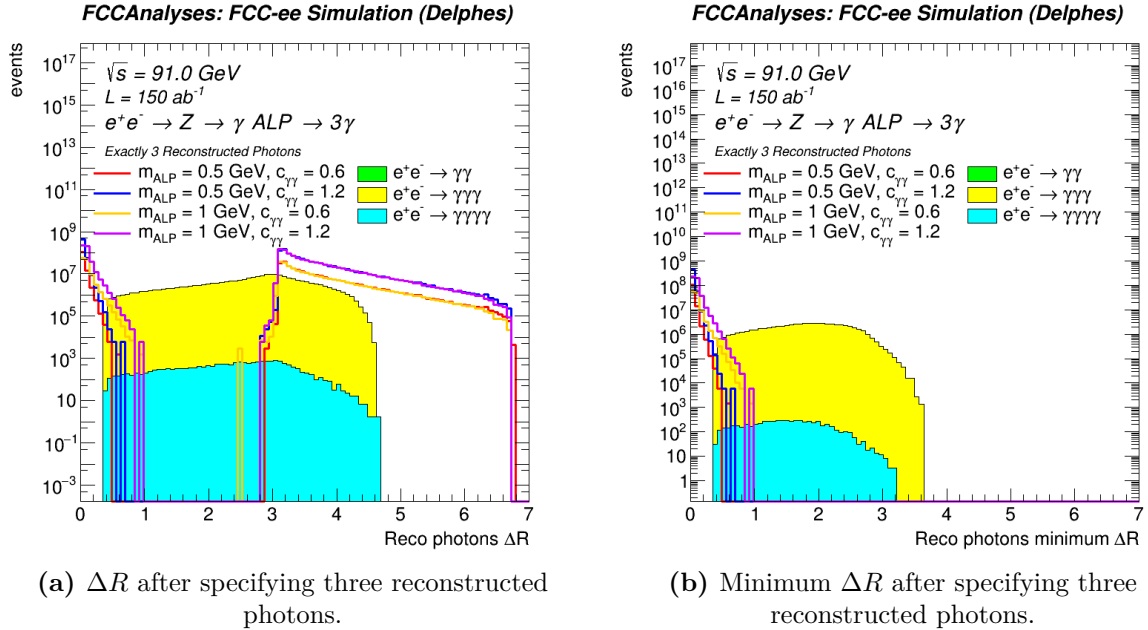
where  $\Delta\eta$  and  $\Delta\phi$  are the difference between the respective angles  $\eta$  (pseudorapidity) and  $\phi$  (azimuthal angle) (shown in Figure 13) for two particles. From this,  $\min(\Delta R)$  is simply determined as the minimum of the previously calculated  $\Delta R$  for each event,



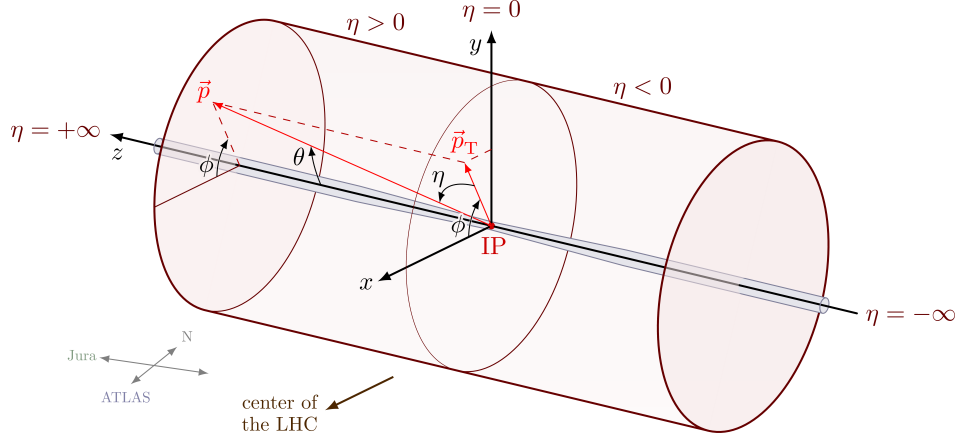
**Figure 11:** “reco” level photons after a selection was made specifying exactly three reconstructed photons.

and the two variables are plotted after applying the selection necessitating three reconstructed photons (Figure 12). Because, as mentioned, the first photon is generated separately from the next two, it follows that  $\min(\Delta R)$  should always represent the angle between the two photons decaying from the ALP.

Analyzing the  $\min(\Delta R)$  plot (Figure 12), the next selection made was to impose the



**Figure 12:**  $\Delta R$  vs.  $\min(\Delta R)$ . Taking the minimum removes the second peak for the signal, proving that  $\min(\Delta R)$  is showing the angle between the two photons decaying from the ALP.



**Figure 13:** Diagram showing how the angles  $\eta$  and  $\phi$  are defined [18].

limitation that  $\min(\Delta R) < 1$ . This was determined, graphically, to be most effective in removing large amounts of background while keeping the vast majority of the signal.

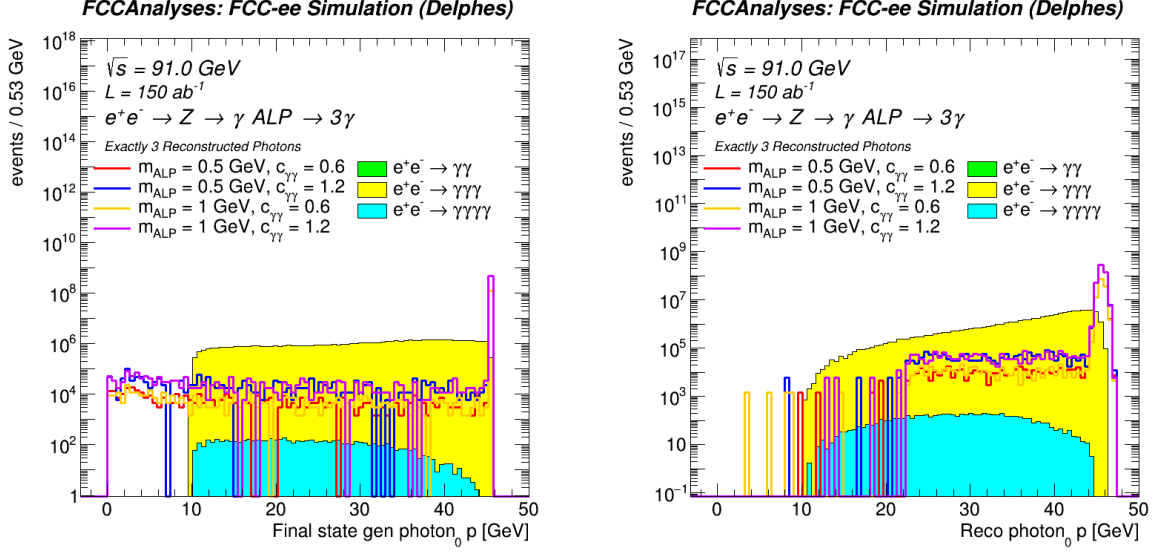
An additional selection that was considered was limiting the momentum of the first photon generated in conjunction with the ALP. As was proven earlier (Figure 8), when the Z boson decays into an ALP and a photon, the ALP will have a momentum of exactly half the center of mass energy  $\sqrt{s}$ ; in our case,  $p_a = \frac{1}{2}(91.0 \text{ GeV}) = 45.5 \text{ GeV}$  (seen in Figure 8). The other half of the center of mass energy will become the momentum of the photon generated along with the ALP (the leading photon, or the photon with the highest transverse momentum  $p_t$ ), meaning the momentum of the leading photon should also be  $45.5 \text{ GeV}$ . Knowing this, and considering the graphical proof for this hypothesis in Figure 14(a) (which shows a peak for the leading photon at the “gen” level at  $45.5 \text{ GeV}$ ), the logical selection is to make sure that at the “reco” level, the momentum of the leading photon is in the correct range (in this case determined to be  $> 44 \text{ GeV}$  in order to keep the most signal).

A summary of these applied selections is shown in Table 2, which lays out both the raw number of events generated ( $n_{raw}$ ) and the expected number of events ( $n$ , calculated from equation (14)) for the four signal ranges and the three backgrounds, both before and after each selection was made. The “Combined” column represents all three aforementioned selections applied together.

From Table 2, it is clear that both the  $\Delta R < 1$  and the photon\_0 momentum  $< 44 \text{ GeV}$  selections were effective, in that they eliminated a significantly larger percentage of the background than the signal, with the former selection taking a slight edge in effectiveness. Additionally, when combining the two selections, the largest amount of background was removed, with only a small amount of additional signal removed. Finally, it is clear that for this analysis at the FCC, the most important background is the tri-photon production background, as it retained by far the highest number of events even after the selections were made. This was, of course, to be expected, as the Feynman diagram for the signal (Figure 2) also results in three final state photons. After all selections were made, 100% of the di- and quad-photon production backgrounds were removed, and 89.66% of the tri-photon production background was removed while maintaining greater than 83% (depending on ALP mass and coupling constant) of the

| Signal                                    |                    | 3 reco photons |           |          |         | $\Delta R < 1$ |          |         |           | $p > 44 \text{ GeV}$ |         |           |          | Combined |           |         |  |
|-------------------------------------------|--------------------|----------------|-----------|----------|---------|----------------|----------|---------|-----------|----------------------|---------|-----------|----------|----------|-----------|---------|--|
|                                           | No sel             |                |           |          |         |                |          |         |           |                      |         |           |          |          |           |         |  |
| $m_a$                                     | $C_{\gamma\gamma}$ | $N_{tot}$      | $n_{raw}$ | $n$      | eff     | $n_{raw}$      | $n$      | eff     | $n_{raw}$ | $n$                  | eff     | $n_{raw}$ | $n$      | eff      | $n_{raw}$ | eff     |  |
| 0.5                                       | 0.6                | 100000         | 84089     | 1.24E+08 | 15.91%  | 84089          | 1.24E+08 | 15.91%  | 83806     | 1.24E+08             | 16.19%  | 83806     | 1.24E+08 | 16.19%   | 83806     | 16.19%  |  |
| 0.5                                       | 1.2                | 100000         | 83843     | 4.96E+08 | 16.16%  | 83843          | 4.96E+08 | 16.16%  | 83539     | 4.94E+08             | 16.46%  | 83539     | 4.94E+08 | 16.46%   | 83539     | 16.46%  |  |
| 1.0                                       | 0.6                | 100000         | 83496     | 1.23E+08 | 16.50%  | 83496          | 1.23E+08 | 16.50%  | 83190     | 1.23E+08             | 16.81%  | 83190     | 1.23E+08 | 16.81%   | 83190     | 16.81%  |  |
| 1.0                                       | 1.2                | 100000         | 83557     | 4.94E+08 | 16.44%  | 83557          | 4.94E+08 | 16.44%  | 83254     | 4.93E+08             | 16.75%  | 83254     | 4.93E+08 | 16.75%   | 83254     | 16.75%  |  |
| Background                                |                    |                |           |          |         |                |          |         |           |                      |         |           |          |          |           |         |  |
| $ee \rightarrow \gamma\gamma$             |                    | 100000         | 0         | 0.00E+00 | 100.00% | 0              | 0.00E+00 | 100.00% | 0         | 0.00E+00             | 100.00% | 0         | 0.00E+00 | 100.00%  | 0         | 100.00% |  |
| $ee \rightarrow \gamma\gamma\gamma$       |                    | 100000         | 97000     | 6.71E+07 | 3.00%   | 13009          | 9.00E+06 | 86.99%  | 13532     | 9.36E+06             | 86.47%  | 10344     | 7.16E+06 | 89.66%   | 10344     | 89.66%  |  |
| $ee \rightarrow \gamma\gamma\gamma\gamma$ |                    | 100000         | 3868      | 6.46E+03 | 96.13%  | 832            | 1.39E+03 | 99.17%  | 4         | 6.68E+00             | 100.00% | 3         | 5.01E+00 | 100.00%  | 3         | 100.00% |  |

**Table 2:** The number of events, for a range of signals and backgrounds, both before and after selections have been made. The “ $\Delta R < 1$ ” and “ $p > 44 \text{ GeV}$ ” columns include the default “3 reco photons” selection, and the “Combined” column combines all three previous selections. “No sel” is simply giving the number of events before any selections were made.  $n_{raw}$  represents the raw number of events for the given ALP mass, coupling constant, and selection (from 100000 total generated events  $N_{tot}$ ),  $n$  represents the expected number of events for the given ALP mass, coupling constant, and selection (calculated from equation (14)), and “eff” represents the percentage of either signal or background which was removed after the respective selection was made.



(a) The momentum of the leading photon at the “gen” level. (b) The momentum of the leading photon at the “reco” level.

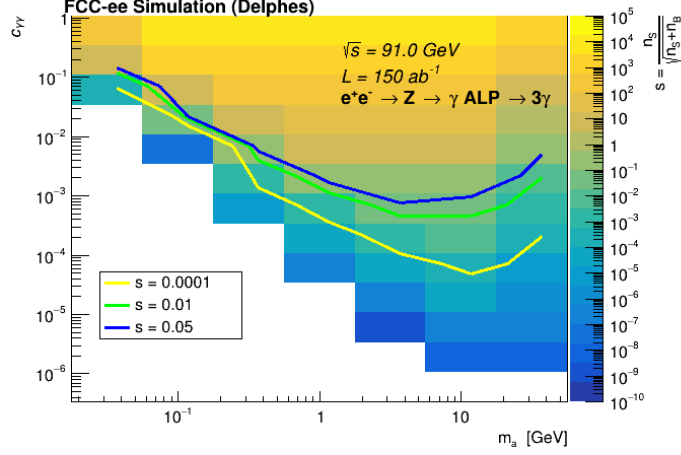
**Figure 14:** The momentum of photon<sub>0</sub> (the photon with index 0, or the leading photon). The selection made from these plots was limiting the momentum at the “reco” level to  $> 44 \text{ GeV}$ .

signal.

#### 4.4 Sensitivity Analysis

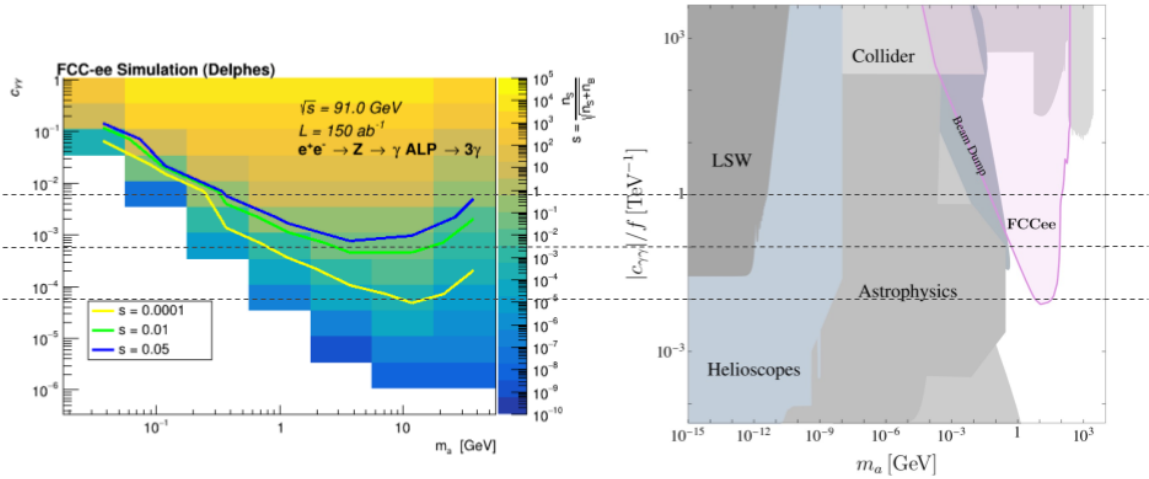
Now that effective selections have been made, the next step is to analyze the sensitivity of the FCC-ee Z pole run given said selections. To do this, many more data points were generated (with 10000 events each) for a much wider range of ALP masses and coupling constants in order to get a broader picture of which values the FCC-ee would be most sensitive to (table with all these values in Appendix C). The ALP masses and coupling constants corresponding to the highest sensitivities would be easiest to detect. The expected number of background events ( $n_B$ ) was calculated using equation (14) and plugging in 10000 for  $n_{raw}$ , and the expected number of signal events ( $n_S$ ) was calculated using equation (14) and plugging in the number of events remaining after the “Combined” selection for  $n_{raw}$ . Then,  $n_B$  and  $n_S$  were utilized in equation (15) to calculate the sensitivity. This sensitivity value was graphed based on both the ALP mass and coupling constant, with logarithmic axes to cover a wider range, in Figure 15. The blue, green, and yellow lines show sensitivity values of 0.01, 0.05, and 0.0001, respectively, visualizing a dip in conjunction with lower coupling constants. The blank squares in the bottom left corner were set to zero because the ALPs at such low masses and coupling constants were so long-lived as to escape the detector entirely. This relationship makes sense based on equation (7).

To further investigate sensitivity at the FCC-ee, the results from this analysis were compared with previously calculated theoretical predictions of FCC-ee sensitivity to the



**Figure 15:** Sensitivity plot visualizing values calculated from equation (14) from ALP mass values 0.1–31.6 and coupling constant values  $6.0\text{E-}07$ – $0.6$ . Squares towards yellow represent higher sensitivity of the FCC-ee Z pole run to the ALP production process in Figure 2.

ALP production process in Figure 2. A plot of these theoretical predictions, alongside the generated plot of simulated sensitivities, is shown in Figure 16 with horizontal lines to align the y-axes. Indeed, when aligning the axes (the theoretical prediction y-axis is the coupling constant divided by  $f_a$ ), the two plots are even more similar in shape. The minimum troughs of the theoretical plot and the yellow line (sensitivity of 0.0001) in the generated sensitivity plot clearly occur at both a similar coupling constant ( $\sim 5\text{E-}04$ ) and a similar mass ( $\sim 10 \text{ GeV}$ ) value, indicating a) that the generation done in this analysis aligns with theoretical predictions, and b) that the selections made in subsection 4.3 were useful in eliminating background while maintaining signal.



**Figure 16:** Generated sensitivity plot alongside theoretical sensitivity plot, with horizontal lines superimposed to better visualize the different y-axes [2].

## 5 Conclusion

This report has built on previous investigations into ALPs, a hypothetical BSM particle, at the FCC-ee. It has focused on long-lived ALPs within a mass range on the  $\text{GeV}$  scale, considering one production process (Figure 2). Using Monte Carlo simulation and the FCCAnalyses framework, relevant processes were generated and analyzed, varying different parameters (mass and coupling constant) of the ALP. Considering a theoretical framework, the lifetime of these ALPs was analyzed, and analytically computed results were compared to Monte Carlo generated results (Table 1). In addition, multiple new variables were added to the analysis scripts of the FCCAnalyses framework, and changes have been made to the code to expand its usefulness. Some of these variables and their relevance have been documented in this report. Finally, background processes with similar final states to the signal were considered, before coming up with useful selections to separate the signal from the background. These selections were analyzed for effectiveness and summarized (Table 2). Using the best of these selections, a sensitivity analysis (Figure 15) was done for a wide range of ALP masses and coupling constants. This report serves as a synopsis of the work done on long-lived ALPs at the FCC-ee.

For future similar projects building on this one, there are many improvements and additions that can be made. The most interesting next step would be to update the FCCAnalyses and EDM4hep code to include calorimeter timing variables on the reconstruction level. With this additional information, one could gain further insight into the ALP production process and complete further analysis of the ALP lifetimes, including considering cases with lifetimes that are either too short or too long to be properly detected. Additionally, this information could lead to potentially new, even more effective selections, which could further remove background events and isolate the relevant signal. Finally, other ALP production processes at the FCC-ee (e.g. generation from a photon or exotic Higgs decays, decay to two leptons or quarks [3]) could be considered to expand the knowledge base surrounding the proposed particle.

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# A Appendix: MadGraph Process Card

MadGraph process card for the ALP signal process (Figure 2), in this case with a mass  $m_a = 1.0 \text{ GeV}$  and a coupling constant  $C_{\gamma\gamma} = 1.0$ . For all signal generation, the mass and coupling constant were varied, and for background generation, a similar process card was used but without importing the model “ALP\_NLO\_UFO” and specifying the background process in place of the signal in the generation line.

```
1 *****
2 **                               MadGraph5_aMC@NLO                               **
3 **                               **                                               **
4 **                               *                               *               **
5 **                               *       *       *       *       *               **
6 **                               * * * * * 5 * * * * *               **
7 **                               *       *       *               **
8 **                               *                               *               **
9 **                               **                                               **
10 **                               **                                               **
11 **                               VERSION 3.2.0                               2021-08-22 **
12 **                               **                                               **
13 **                               The MadGraph5_aMC@NLO Development Team - Find us at **
14 **                               https://server06.fynu.ucl.ac.be/projects/madgraph **
15 **                               **                                               **
16 *****
17 **                               **                                               **
18 **                               Command File for MadGraph5_aMC@NLO               **
19 **                               **                                               **
20 **                               run as ./bin/mg5_aMC filename                   **
21 **                               **                                               **
22 *****
23 set group_subprocesses Auto
24 set ignore_six_quark_processes False
25 set low_mem_multicore_nlo_generation False
26 set complex_mass_scheme False
27 set include_lepton_initiated_processes False
28 set gauge unitary
29 set loop_optimized_output True
30 set loop_color_flows False
31 set max_npoint_for_channel 0
32 set default_unset_couplings 99
33 set max_t_for_channel 99
34 set zerowidth_tchannel True
35 set nlo_mixed_expansion True
36 set auto_convert_model T
37 import model sm
38 define p = g u c d s u~ c~ d~ s~
39 define j = g u c d s u~ c~ d~ s~
40 define l+ = e+ mu+
41 define l- = e- mu-
42 define vl = ve vm vt
43 define vl~ = ve~ vm~ vt~
44 import model ALP_NLO_UFO
45 generate e+ e- > Z, (Z > a ALP, (ALP > a a))
46 output madgraph_1.0_1.0
47 launch madgraph_1.0_1.0
48 # shower = Pythia8
```

```

49 # detector = Delphes
50 # analysis = OFF
51 done
52 # set to electron beams (0 for ele, 1 for proton)
53 set lpp1 0
54 set lpp2 0
55 set ebeam1 45.594
56 set ebeam2 45.594
57 # set ALP mass
58 set Ma 1.0
59 # set ALP couplings
60 set cWW = 0.0
61 set CYY = 1.0
62 set cGG = 0.
63 set cuu = 0.
64 set cdd = 0.
65 set ccc = 0.
66 set css = 0.
67 set ctt = 0.
68 set cbb = 0.
69 set cee = 0.
70 set cmumu = 0.
71 set ctautau = 0.
72 set cah = 0.
73 set cZh5 = 0.
74 # set supression scale in the effective operators coupling the ALP to SM
    particles: 1000/(16*pi**2)
75 set falp = 6.33
76
77 # set the decay width of the ALP to auto
78 set WALP auto
79 set time_of_flight 0
80 set nevents 100000
81 done

```

## B Appendix: Pythia Process Card

Pythia process card, responsible for dictating the parameters regarding showers/hadronization as well as Delphes reconstruction. The output file from the MadGraph generation is unzipped and used as input here. For all samples used in this report, either 100000 or 10000 events were generated. The path under “Beams:” was altered depending on the generated sample.

```
1 ! File: ALP_pythia.cmd
2 Random:setSeed = on
3 Main:timesAllowErrors = 10          ! how many aborts before run stops
4 Main:numberOfEvents = 100000       ! number of events to generate
5
6
7 ! 2) Settings related to output in init(), next() and stat().
8 Next:numberCount = 100             ! print message every n events
9 !Beams:idA = 11                    ! first beam, e+ = 11
10 !Beams:idB = -11                   ! second beam, e- = -11
11
12 Beams:frameType = 4                 ! read info from a LHEF
13 ! Change the LHE file here
14 Beams:LHEF = /eos/experiment/fcc/ee/analyses_storage/BSM/LLPs/
    ALPs_3photons/mg5amcnlo/madgraph_1.0_1.0/Events/run_01/
    unweighted_events.lhe
15
16 ! 3) Settings for the event generation process in the Pythia8 library.
17 PartonLevel:ISR = on                ! initial-state radiation
18 PartonLevel:FSR = on                ! final-state radiation
19
20 LesHouches:setLifetime = 2
```

## C Appendix: Table of Values for Sensitivity Plot

A table containing all the generated values, for reference, used to create the sensitivity plot (Figure 15).  $\sigma$  represents the cross-section for each respective ALP mass and coupling constant,  $n_{raw}$  represents the raw number of events after all selections have been made (total number of generated events is constant at 10000 events, except for the backgrounds, with 100000 generated events),  $n$  represents the expected number of events (signal or background) calculated from equation (14), and  $s$  is the sensitivity calculated from equation (15).

| $m_a$ (GeV) | $C_{\gamma\gamma}$ | $\sigma$ (pb <sup>-1</sup> ) | $n_{raw}$  | $n$      | $s$      |
|-------------|--------------------|------------------------------|------------|----------|----------|
| Signal      |                    |                              |            |          |          |
| 0.0316      | 0.000019           | 9.9889E-10                   | 0          | 0.00E+00 | 0.00E+00 |
| 0.0316      | 0.00006            | 9.8610E-09                   | 0          | 0.00E+00 | 0.00E+00 |
| 0.0316      | 0.00019            | 9.8890E-08                   | 0          | 0.00E+00 | 0.00E+00 |
| 0.0316      | 0.0006             | 9.8620E-07                   | 0          | 0.00E+00 | 0.00E+00 |
| 0.0316      | 0.0019             | 9.8890E-06                   | 0          | 0.00E+00 | 0.00E+00 |
| 0.0316      | 0.006              | 9.8620E-05                   | 0          | 0.00E+00 | 0.00E+00 |
| 0.0316      | 0.019              | 9.8890E-04                   | 0          | 0.00E+00 | 0.00E+00 |
| 0.0316      | 0.06               | 9.8610E-03                   | 0.009889   | 1.46E+00 | 5.47E-04 |
| 0.0316      | 0.19               | 9.8890E-02                   | 9          | 1.34E+04 | 4.98E+00 |
| 0.0316      | 0.6                | 9.8610E-01                   | 62         | 9.17E+05 | 3.23E+02 |
| 0.1         | 0.000019           | 9.8890E-10                   | 0          | 0.00E+00 | 0.00E+00 |
| 0.1         | 0.00006            | 9.8610E-09                   | 0          | 0.00E+00 | 0.00E+00 |
| 0.1         | 0.00019            | 9.8890E-08                   | 0          | 0.00E+00 | 0.00E+00 |
| 0.1         | 0.0006             | 9.8610E-07                   | 0          | 0.00E+00 | 0.00E+00 |
| 0.1         | 0.0019             | 9.8890E-06                   | 0          | 0.00E+00 | 0.00E+00 |
| 0.1         | 0.006              | 9.8610E-05                   | 0.00009861 | 1.46E-04 | 5.45E-08 |
| 0.1         | 0.019              | 9.8890E-04                   | 14         | 2.08E+02 | 7.76E-02 |
| 0.1         | 0.06               | 9.8610E-03                   | 168        | 2.48E+04 | 9.27E+00 |
| 0.1         | 0.19               | 9.8890E-02                   | 1358       | 2.01E+06 | 6.65E+02 |
| 0.1         | 0.6                | 9.8610E-01                   | 3645       | 5.39E+07 | 6.90E+03 |
| 0.316       | 0.000019           | 9.8880E-10                   | 0          | 0.00E+00 | 0.00E+00 |
| 0.316       | 0.00006            | 9.8610E-09                   | 0          | 0.00E+00 | 0.00E+00 |
| 0.316       | 0.00019            | 9.8880E-08                   | 0          | 0.00E+00 | 0.00E+00 |
| 0.316       | 0.0006             | 9.8610E-07                   | 2          | 2.96E-02 | 1.11E-05 |
| 0.316       | 0.0019             | 9.8880E-06                   | 28         | 4.15E+00 | 1.55E-03 |
| 0.316       | 0.006              | 9.8610E-05                   | 378        | 5.59E+02 | 2.09E-01 |
| 0.316       | 0.019              | 9.8880E-04                   | 3111       | 4.61E+04 | 1.72E+01 |
| 0.316       | 0.06               | 9.8610E-03                   | 7374       | 1.09E+06 | 3.80E+02 |
| 0.316       | 0.19               | 9.8880E-02                   | 7722       | 1.15E+07 | 2.65E+03 |
| 0.316       | 0.6                | 9.8610E-01                   | 7764       | 1.15E+08 | 1.04E+04 |
| 1           | 0.000019           | 9.8850E-10                   | 0          | 0.00E+00 | 0.00E+00 |
| 1           | 0.00006            | 9.8580E-09                   | 6          | 8.87E-04 | 3.32E-07 |
| 1           | 0.00019            | 9.8850E-08                   | 43         | 6.38E-02 | 2.38E-05 |
| 1           | 0.0006             | 9.8580E-07                   | 555        | 8.21E+00 | 3.07E-03 |
| 1           | 0.0019             | 9.8850E-06                   | 4280       | 6.35E+02 | 2.37E-01 |
| 1           | 0.006              | 9.8580E-05                   | 8342       | 1.23E+04 | 4.61E+00 |
| 1           | 0.019              | 9.8850E-04                   | 8352       | 1.24E+05 | 4.59E+01 |
| 1           | 0.06               | 9.8580E-03                   | 8340       | 1.23E+06 | 4.26E+02 |
| 1           | 0.19               | 9.8850E-02                   | 8353       | 1.24E+07 | 2.80E+03 |
| 1           | 0.6                | 9.8580E-01                   | 8333       | 1.23E+08 | 1.08E+04 |
| 3.16        | 0.0000019          | 9.8530E-12                   | 8151       | 1.20E-03 | 4.50E-07 |
| 3.16        | 0.000006           | 9.8260E-11                   | 4          | 5.90E-06 | 2.20E-09 |

|                                           |           |            |       |          |          |
|-------------------------------------------|-----------|------------|-------|----------|----------|
| 3.16                                      | 0.000019  | 9.8530E-10 | 59    | 8.72E-04 | 3.26E-07 |
| 3.16                                      | 0.00006   | 9.8620E-09 | 651   | 9.63E-02 | 3.60E-05 |
| 3.16                                      | 0.00019   | 9.8530E-08 | 4468  | 6.60E+00 | 2.47E-03 |
| 3.16                                      | 0.0006    | 9.8260E-07 | 8179  | 1.21E+02 | 4.51E-02 |
| 3.16                                      | 0.0019    | 9.8530E-06 | 8140  | 1.20E+03 | 4.50E-01 |
| 3.16                                      | 0.006     | 9.8260E-05 | 8147  | 1.20E+04 | 4.48E+00 |
| 3.16                                      | 0.019     | 9.8530E-04 | 8164  | 1.21E+05 | 4.47E+01 |
| 3.16                                      | 0.06      | 9.8260E-03 | 8128  | 1.20E+06 | 4.14E+02 |
| 3.16                                      | 0.19      | 9.8530E-02 | 8181  | 1.21E+07 | 2.76E+03 |
| 3.16                                      | 0.6       | 9.8260E-01 | 8143  | 1.20E+08 | 1.06E+04 |
| 10                                        | 0.0000006 | 9.5100E-13 | 6     | 8.56E-08 | 3.20E-11 |
| 10                                        | 0.0000019 | 9.5360E-12 | 77    | 1.10E-05 | 4.12E-09 |
| 10                                        | 0.000006  | 9.5100E-11 | 623   | 8.89E-04 | 3.32E-07 |
| 10                                        | 0.000019  | 9.5360E-10 | 4155  | 5.94E-02 | 2.22E-05 |
| 10                                        | 0.00006   | 9.5100E-09 | 6624  | 9.45E-01 | 3.53E-04 |
| 10                                        | 0.00019   | 9.5360E-08 | 6239  | 8.92E+00 | 3.34E-03 |
| 10                                        | 0.0006    | 9.5100E-07 | 6186  | 8.82E+01 | 3.30E-02 |
| 10                                        | 0.0019    | 9.5360E-06 | 6197  | 8.86E+02 | 3.31E-01 |
| 10                                        | 0.006     | 9.5100E-05 | 6201  | 8.85E+03 | 3.30E+00 |
| 10                                        | 0.019     | 9.5360E-04 | 6202  | 8.87E+04 | 3.30E+01 |
| 10                                        | 0.06      | 9.5100E-03 | 6195  | 8.84E+05 | 3.12E+02 |
| 10                                        | 0.19      | 9.5360E-02 | 6170  | 8.83E+06 | 2.21E+03 |
| 10                                        | 0.6       | 9.5100E-01 | 6186  | 8.82E+07 | 9.03E+03 |
| 31.6                                      | 0.0000006 | 6.7180E-13 | 18    | 1.81E-07 | 6.78E-11 |
| 31.6                                      | 0.0000019 | 6.7370E-12 | 114   | 1.15E-05 | 4.31E-09 |
| 31.6                                      | 0.000006  | 6.7180E-11 | 298   | 3.00E-04 | 1.12E-07 |
| 31.6                                      | 0.000019  | 6.7370E-10 | 354   | 3.58E-03 | 1.34E-06 |
| 31.6                                      | 0.00006   | 6.7180E-09 | 354   | 3.57E-02 | 1.33E-05 |
| 31.6                                      | 0.00019   | 6.7370E-08 | 350   | 3.54E-01 | 1.32E-04 |
| 31.6                                      | 0.0006    | 6.7180E-07 | 351   | 3.54E+00 | 1.32E-03 |
| 31.6                                      | 0.0019    | 6.7370E-06 | 350   | 3.54E+01 | 1.32E-02 |
| 31.6                                      | 0.006     | 6.7180E-05 | 357   | 3.60E+02 | 1.34E-01 |
| 31.6                                      | 0.019     | 6.7370E-04 | 348   | 3.52E+03 | 1.31E+00 |
| 31.6                                      | 0.06      | 6.7180E-03 | 349   | 3.52E+04 | 1.31E+01 |
| 31.6                                      | 0.19      | 6.7370E-02 | 354   | 3.58E+05 | 1.30E+02 |
| 31.6                                      | 0.6       | 6.7180E-01 | 350   | 3.53E+06 | 1.08E+03 |
| <b>Background</b>                         |           |            |       |          |          |
| $ee \rightarrow \gamma\gamma$             |           | 5.7318E+01 | 0     | 0.00E+00 | N/A      |
| $ee \rightarrow \gamma\gamma\gamma$       |           | 4.6135E-01 | 10344 | 7.16E+06 | N/A      |
| $ee \rightarrow \gamma\gamma\gamma\gamma$ |           | 1.1138E-03 | 832   | 1.39E+03 | N/A      |